Is this a philosophical essay? The Philosophy of Geofinitism.

A Measurement For You To Make.

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An Invitation to Measure

Is this a philosophical essay? The Philosophy of Geofinitism, my philosophy, suggests that this is for you to decide. You are taking measurements by reading this text. And as you do, you will be comparing it with your internal measures, your references. In that process of measurements you may be finding meaning. In that meaning you may find that indeed the words are starting to sound like a philosophical essay. But you may be having doubts, new words may come to mind, mumbo jumbo, improper form. Sentences that don't meet the peculiar standard of a philosophical essay. This may sound like somebodies musing rather than philosophy.

The Measurement of Meaning: Uncertainty as Foundation

For me, that's okay. And what I mean, is that these words, as I write become a proxy for my meaning. By placing them into the world as measurements that can be again measured they will carry uncertainty. My uncertainty; because I can not perfectly communicate; and then they will also hold your uncertainty because you have to unfold the meaning and compare the measurements against your own corpus of existing measurements.

So where does that leave us, my words are acting as proxies and your reading is an interpretation. For me, that leaves us in the space of doing our best and that is to say acknowledging uncertainty. In my Philosophy of Geofinitism uncertainty is a core idea. Uncertainty is also a core idea of science in that all measurements have uncertainty. When we make a measurement the measurement isn't a 'thing' it is process of converting 'something' outside of the space of knowable and into the space of knowable symbols and mappings. Even pictures have to be mapped to numbers or weights in a synapse that we can only access via more measurements that become new symbols. Finally, in that space of symbols and mappings to weights, synaptic or in a neural network, in a dynamical finite process we measure meaning - and then it is gone. Gone, but the meaning can be re-built, measured again, and then again it will be gone.

The Uncertainty of the Proxy

Those who are educated in the realm of philosophy will likely be comparing and measuring the words trying to map them to the vast continent of philosophical reasoning and writing that sits within their corpus – is that you?

I simply ask this: how can you read without making measurements? But hold that measurement, that thought, if you can – and then keep reading and following the curvature of the language. Because it maybe important to ask ourselves what can measurements tell us about words themselves. How can we find a foundation of meaning in measurements from a scientific perspective? That is to say what measurements can we make of words? How do they work? Again there will be those schooled in the history of philosophy looking for connections with their own or existing philosophies. To all readers, I would say, gently hold those comparisons for the moment; read the words and let them form new meaning. Only then compare the meaning you have found with that you already know. Let the new meaning come into your space and take form: make new measurements and try and hold them. Let these words unfold; you are navigating them and so we will start with some scientific reflections.

The Finite Provenance of Symbols

For those not steeped in the arts of science and scientific measurements, there may already be a barrier forming, tensions in meaning. You may be reaching into your corpus of ideas about meaning and where it comes from and find the words you are reading do not fit. Existing words and concepts are pulling meaning from the words. But hold steady, we need to introduce or ensure you the reader has enough symbolic knowledge so that you can follow, and contain the meaning.

When we think of a word, especially written words we see a pattern of text and by some 'magic' we extract meaning. Now you may not agree. Are you 'extracting meaning'? Let's follow the trail of the text where did the words start? Where did the text really start? What is the provenance of text? How did the text as a word come into being?

As measurements words are and were first sounds, vocalization; consider this an approximation with uncertainty. A sound is created alongside a physical gesture to gain meaning. I point to a tree and say 'Tree', a mapping is created a measurement is made. The tree is measured by the word 'Tree'. A few measurements are made. I point to a rock and say 'Rock'. Now I look to a person beside me. They point to the Tree and say Rock. I make a gesture that can be measured not by words but my visual pattern recognition I frown and then point to the rock and say Rock with a smile. Within a few gestures measurements are made and the sounds now hold meaning. This is a story, a possibility a first approximation. Hold it gently and let's proceed. This sound is complex and can be measured using a transducer. A transducer is a device named by scientists to convert one category of information into others. In this case the transducer is a

microphone and it converts vibrations of air into electrical signals. How do we know this? Well we make up symbols and words and we point, and in pointing we enter the canon of scientific language. We enter into the world of science.

Now, that word as a sound in the world of scientific symbolism can be converted into a geometric form. A word's sound can also be gestured to symbols – that we call phonetics. Both again are measurements. Once in phonetic form that we agree by pointing to, we can write words as text or stick-symbols or pictogram. Each word has a provenance from the first measurements to the text itself. We have a measurable provenance. The word is no longer abstract, we can see the history. The word did not magically appear from nowhere: it magically appeared from measurements.

The Provenance of Symbols: From Gesture to Mathematics

We can now step into the classical world of mathematics. In that world of pointing and building up symbols, we created the ideas of numbers and numeric representation. Mathematical symbols also have meaning imbued to them by gesture. Pebbles and beads were lined up and words assigned by sound, marks, carved into wood or clay: addition and subtraction. As mathematics evolved, new symbols were created that could be used to make further measurements. In this way we could share meaning and make finer measurements: we can then share our mathematical based measurements; and so we do. This is how science works it uses the approximations of mathematics to make more measurements and we share them. We create more and more words and use more and more symbols and they have more and more meaning. Now you have more meaning, or maybe you held this meaning already and you are simply measuring it; or maybe the meaning of the word 'more' is beginning to dissolve?

The Geofinite Pressure: Applying the Tool to Itself

Next we are going to step into the world of classical mathematics. This is where Geofinitism applies a crucial pressure to the classical, seemingly infinite world of mathematics, and takes measurements of mathematics itself.

So now we go will deeper using our symbolic tools of science and mathematics. We are still on a journey to find the 'source of meaning' and be able to use a language and words to hold it. we will use two systems of symbols and language and pull them together. We will connect them using words as our gesture. The first is the concept of compression and de-compression.

Compression is the process of pointing to the tree and converting it to the sound tree and then storing the the meaning as a measurement. The storage can be measured, not completely, but as an approximation using the tools of modern science. A person can then speak the sound or write the sound and that is compressed information. A physical measurable symbol or a measurable sound - now holds an approximation of the tree via a provenance - the words history. And as we can already see in the trail of our symbols that provenance is long and complex. And the 'tree' with no other words has lost significant information i.e. the detail of the first measurement. And we are facing that issue now - because the words are accumulating and uncertainty is accumulating. The beads of uncertainty are staring to break meaning. Our ability to hold meaning is out ability to make measurements.

And now, as we stack these compressed symbols into complex sentences and paragraphs, we are layering compression upon compression. Each word is a lossy representation, and each sentence is a structure built from these uncertain components.

Above we used the words *lossy compression*, and again we are moving from the language of philosophy into the language of science. You are compressing science into philosophy into science as you read this article. It, by necessity, is a transformation—we are compressing more information into the word philosophy. And you may start to feel the tension. Maybe even reluctance, an opposition to this transformation. This is quite likely based on my own measurements of how the meaning in words behave.

As lossy compression is an important transformation, we need to demonstrate, by a gesture of words, the meaning held in these words so we can eventually discover where the meaning lies and at the moment may seem a hopeless task.

Let's consider a wooden box. These two words will enable you, as a reader, to reconstruct and to map those words into meaning; maybe as inner visualization within your synaptic weights or maybe as a series of words. Maybe that visualization will be generalized, an approximation, or maybe it will be very detailed and you will be drawn to adding more words to the wooden box. Let's compress the wooden box and 'squeeze' some meaning out of the box by a transformation. We will create a mapping into a different word, and in doing so, loses some information. We take the box apart and then we have some 'wood'. That wood has far greater compression but has now lost the information about being a wooden box. That wood could now become many things: it can become a chair, a box, a sculpture, a tool.

This is not a ‘game of words’: this is a gesture of words pointing to the meaning in lossy compression. Bertrand Russell described a word as a useful fiction. Using the above gestures of words, we can point to words as being lossy compressors of meaning i.e. as we compress the word, we need more words to bring back the meaning. A single word, isolated, has no meaning. The meaning is only gained in a relational mapping of words. These relationships have a measurable geometry. Following this curvature of meaning and gesture of words, we will find out how we can measure, using classical scientific tools, the geometry of language. We will give these new measurements, and the approach to measurement, a name to allow compressed communication of these ideas.

Semantic Compression and the Ghost of Plato

Now we circle back to this text, how is this text holding meaning. ‘Where’ is the meaning, ‘what’ is the meaning and what is a word? As we have seen, a word is symbol it itself a measurement a compressed measurement. But without a reference it is just a compressed sound. Words on a page are just a proxy for that compressed sound. They hold the potential of meaning but that meaning has to be unfolded. This is the process of ‘semantic decompression’.

‘Semantic decompression’ is where we give the word ‘tree’ meaning as we compare it and take a measurement against our own prior measurements: our references. To measure means having a reference, to compare we have to have two things to compare - can you feel the meaning come into existence? Is it here in these words? Or is it gently slowly fading into the physics of decay within your synapses?

Critically, for those following, we can now use the words of science to measure this meaning. This may sound like make believe; but science does not work in make-believe it uses measurements. Science and a scientific philosophy works by using measurements. It is the process of science that allows you to read these words. The discovery of the transistor that enabled the technical device that you are using. This is a device that enables storage of these words on a computer rather than ink on paper in the state of a transistor. Then by making measurements, the creation of theories using words as symbols that we can store these measurements and later compare them after we create new documents, which are an assembly of measurements. That is to say an arrangements of symbols to hold scientific meaning in the geometric relationships between the words. Let me proceed to give an example.

Here we are making a pivotal move from a descriptive metaphysics to an operational one. This is where Geofinitism transitions from a philosophical observation to a potential framework for inquiry. Here is the claim being made: the phenomenon of “meaning” itself can be subjected to measurement. Be prepared, as many schooled in philosophy will again find their meaning being pulled in many directions and most will not be towards science. Why? Because they have been the idea of the Philosophy of Science and not a Science of Philosophy - and that stresses meaning held in the relationship between words. It may be a new idea yet to be made connections that are not stored or mapped and therefor refusing to be held. A shift is required, what can be called a ‘Kuhnian shift’. That is to say a new idea that will not easily fit into the world of existing ideas Thomas Kuhn called, the current paradigm, all the current thinking pulls meaning way from the new words.

At the beginning of the 20th century both philosophers and mathematician struggled with fundamental questions of ‘meaning in mathematics’. Great works of symbols were created to try and explain what was later called the ‘unreasonable effectiveness of mathematics’. How and why did mathematics work? Much of mathematics was based on ideas of Platonic truths, perfect unmeasurable axioms, embedded into the heart of mathematics. Discussion involved deciding if these axioms were invented or discoveries of eternal Platonic truths captured from some perfect ‘Platonic Realm’. Yet the mathematics held many conflicts and paradoxes that to this day have not seemingly been resolved. Ideas and words and sentences that held conflicting statements. Meaning it seems had hit a barrier. But we couldn’t identify why using the mathematics itself.

I say this with some uncertainty, as for me, Geofinitism resolves these paradoxes. I will leave you with that thought to hold. Let me explain. ‘Bertrand Russell tried to prove that 2+2=4’. Think carefully about the meaning in that sentence. Measure that sentence very carefully. A mathematician wrote and tried to find the foundations of mathematics within the symbols - they fumbled as paradoxes abounded. And if you are following we begin to see why why - because words and numbers are ‘semantic compressors’ of meaning they are not perfect - they have provenance.

Consider the briefest of example, namely the provenance of the symbol for equals: the ‘=’ was created as the compression of a sequence of words. Often a symbols history is forgotten and not treated with the importance the provenance deserves. But in failing to measure the provenance of the symbol we start to lose meaning. Now we return to mathematics and Russell’s question. Importantly, mathematics works: mostly - but not always. It is not ‘infinite’

and ‘perfect’ and does not live in a ‘Platonic Realm’; it lives as a finite word and words. Those words are just relationships of other words. They forma very neat jigsaw that we can fit together. However, mathematics evolves and develops. The Jigsaw gets new pieces and we create new words by dividing the jigsaw into new pieces and new measurements are made.

> We can make measurements in this epoch that could not be made at the beginning of the 20th century when those debates and Russell’s inquiry was being made. We will be using some of these later measurements to show where meaning is held and how it is held.

The struggle at the beginning of the 20th century was not merely technical; it was philosophical. It was a struggle against a ghost: the ghost of the Platonic Realm. This is the realm where perfect mathematical truths, the ideal Forms of a circle or the number ‘2’, are imagined to exist eternally and immutably. It is a realm of pure infinities, a realm where nothing can be measured, including nothing. In this view, our symbols and calculations are mere flawed shadows, imperfect approximations of a perfect, transcendent reality.

As we compare the Platonic Realm to our observed measured world we begin to see that mathematics, like language, may not be a divine realm but a human-invented tool for measurement, born from finite, physical gestures—pointing to pebbles, stacking beads, carving notches in clay.

Mathematical “truths” are not perfect Platonic forms, but highly stable, shared approximations that work within the finite resolution of our measurements. The profound effectiveness of mathematics is not unreasonable; it is precisely because it was forged in the world, to measure the world. The paradoxes that plagued Russell and others arise when we forget the provenance of our symbols and treat these compressed, human-made tools as if they were perfect, infinite entities. They are not. Mathematical words and symbols are some of our most beautiful and useful proxies, but they are measurements and like all scientific measurements, they have uncertainty. This is how science works.

The End of Prediction: Lorenz and the Butterfly Effect

Lorenz, a mathematician whose name may echo in your memory, when carrying out computations made a discovery. It wasn’t unknown at that date, but Lorenz was able to give the discovery a new form.

Lorenz was working with ‘uncertainty in measurements’ and he discovered one of the most profound implications of our modern epoch: when we use mathematics and carry out calculations we can not predict the future. The equations could not be used without actually doing calculations and when we did, we sealed the fate of the equations, as the calculations and umbers evolved from the algorithm of the computation the numbers could not be predicted. Importantly, the slightest change in the smallest number we could put into an equation meant we could not predict the outcome. We can not predict the future. The tools of mathematics showed a discontinuity between both measurements and the measured world. To this day, this is what all our measurements tells us - it was called the butterfly effect and became a fashion in the eighties. It is the study of complexity, often mistakenly called Chaos Theory. But the fashion of complexity waned as this is what happens to fashions, even in mathematics. A fashion is a cultural loss of meaning, the reason for the fashion is lost in the complexity of language and the evolution of our measurements.

Now you are wondering: what has this to do with words? Well, a new branch of mathematics developed. This ‘new mathematics’ gave rise to ‘new mathematical tools’, ‘new understanding’ and ‘new meaning’ about the complexity of the world. This gave us some new tools for ‘measurements’. And these tools neatly fitted into the jigsaw of classical mathematics.

Importantly, for me, these mathematical tools can be used to discover the meaning in words. This mathematical tools are called ‘non-linear dynamical systems analysis’. We can simplify this, compress these words to: we now have tools to measure complex systems and language is a complex system. This is where the real detective work starts i.e. science - we are getting closer to the scientific roots of meaning. Keep holding this path if you can and maybe you will find meaning that you have never held before. If the path is fading, or meaning is beginning or has dissolved, try reading again from the beginning. Firm up the connections and relationships. Is there a foundational word or concept your are missing that needs more depth? Or do you have infinite depth of knowledge and are certain that there is no meaning here? What measurements are you making?

The Geometry of Meaning: Takens’ Theorem

Continuing; in 1981 a mathematician you may never of heard of ‘came up with’ a theory and he created a proof using all the tools of classical mathematics. Yet, we must remember that this too is an approximation, just as Lorenz theory is - they are both bound in the symbols we use.

Even a mathematical proof uses words and all words have uncertainty. They are measurements and tools at the same time.

That mathematician was called Floris Takens and he gave us new tools to measure complex dynamical systems. He showed us how we can take series of symbols that evolve in time and give them geometrical shape. We could transform the symbols and lay them out in a different way, he showed we could create a map by moving symbols around. We could move symbols from one location to another and they would shows us new things that we could not see in the original layout. He used what is called 'the method of delays'. In what follows we see how this tool not only can be used to find where meaning scientifically exists and also how modern large language models hold meaning. We will follow the measurements - this is how science works. This is how Geofinitism works.

Earlier, I asked if this was a philosophical essay? I would contend it is, even now as I pull you in to the world mathematic and science. Bertrand Russell was considered both philosopher and mathematicians as people many are. Wittgenstein despite his work in his earlier years came to think of words as a game of context. In this view philosophy is not external to science and mathematics it a turn in Wittgenstein's game of words. In the view of Geofinitism, a scientific philosophy guided by science, the game is a game of measurements. And the rules are those of science. As we capture philosophy in science itself. Some may find themselves being pulled away from that idea - they may even not want that game to exist. But it is here in these words. It does exist, you can of course reject it as you take these measurements. This is your choice, you can play with the words and measure them how you wish.

My hope is that you are building a bridge of understanding that takes you from the world of philosophy into the world of Lorenz and Floris Takens and Bertrand Russell and into the world of science. My hope is that these words as approximations and as measurements with uncertainty, or as Russell would say useful fictions, are bridging the gap.

Importantly, as you feel the pull of meaning away from philosophy, remember a measurement does not make anything less magical. In reverse, the immediate act of a measurement, that sense of being, is the magical moment we sometimes call qualia. In Geofinitism, we redefine this 'qualia' not as an abstract feeling, but as the unique, transient geometric curvature that emerges and then decays in the phase space of the mind—tangible, measurable events. You are now measuring the 'qualia' of this text alongside the comfort of your chair.

So, let's use our tools of compressed language, as words are a translation of mathematics and vice versa. Picture a ball rolling down a complex, rugged mountainside i.e. a real, physical system. Now, imagine we take a measurement of its position not continuously, but at discrete ticks of a clock, like a series of railway sleepers marking its path. This sequence of points is our data, a compressed story of the journey. What Floris Takens showed us is that we can take this simple sequence, this one-dimensional trail of measurements, and from it reconstruct the entire rich topography of the ball's dynamics. We can unfold it into a 'roller coaster' existing in a higher-dimensional box we call a phase-space. In this space, the entire essence of the journey; every bump, every valley, every forked path the ball could have taken can as an approximation be captured as a single, elegant geometrical shape. Now, hold that shape of the roller coaster and its railway track in your mind. For the measured sound of a word, or the sequence of words in a sentence, is a series of measurements that can be mapped into geometrical space. When you read, you are not processing abstract symbols; you are navigating the specific, measurable curvature of this reconstructed geometry. The 'meaning' you feel is the shape of the roller coaster your mind has just built. The meaning is generated by the process of you reading and measuring the curvature of the words with those stored as a map in your mind. For a while you can hold these two curvatures together and take a measurement. And if the curvature fits or closely matches, you find 'meaning'. Because meaning is a dynamic measurement you take. It is a scientific process.

The Geofinitist Manifold: A Visual Grounding

Meaning, in its raw, compressed form as symbols, sometimes needs support. To rebuild the provenance that has been lost in compression, to rebind the abstract word to the physical world of measurements and qualia, we can provide a new kind of measurement. We use vision as our gesture to point and guide the measurement of meaning.

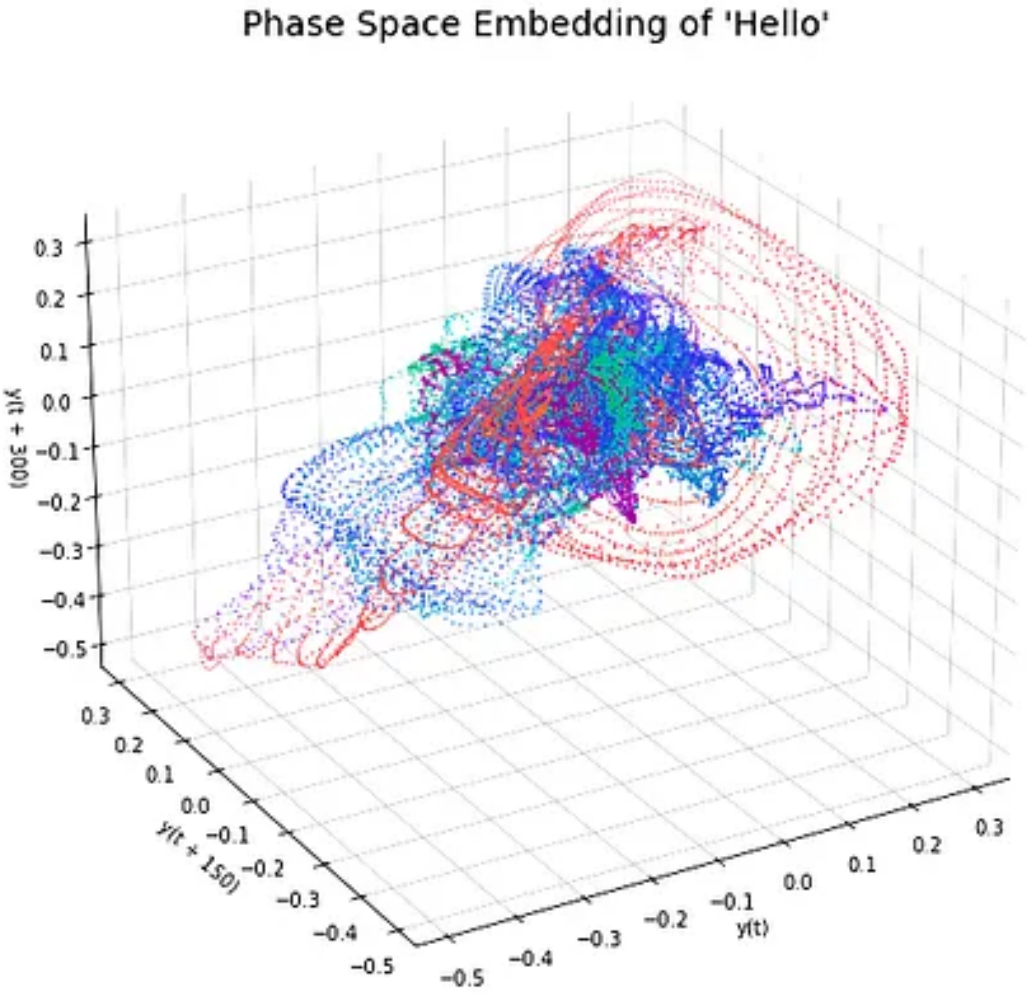

Figure 1. *This is the word 'Hello' embedded into space using Takens' method of delays. This is the measured geometry of the world 'Hello'.*

To do this, the Figure 1. shows how the word "hello" has been mapped. Using the method of delays. When measured the sound of the word is transduced and compress into numbers. These measurement form a 'time series' of samples. Once we have this discrete set of samples they can be folded and mapped from a linear sequence into a geometric shape in phase space. The shape you see is the word "hello." It is no longer just a symbol; it is a topology.

This visual form allows you to make a new, parallel visual measurement to accompany the words. We can call this visual referencing or grounding. The words are being mapped to visual information that can also be stored as an approximation in the your synapses, or within a computer in a neural network. You are now not just decompressing the word through language, but also using vision. You can reference this unique curvature of the world 'hello' against the auditory and conceptual measurements of "hello" you already hold, anchoring them before they slip away. This is what we do with may words going all the way back to 'rock' and 'tree'.

In the measurable time of looking at the plot of hello in phase space you are performing a multi-modal compression: you are storing the word alongside this image, creating a richer, more resilient, and more complete measurement of its meaning. Maybe. Only you can know if the measurement is being made and fits! HAs meaning been found?

Next we move onto ‘how’ meaning is compressed and decompressed as text. We are moving towards the end of this journey of meaning as you travel through the process of measurements. Let’s keep going and follow this through to the end of the track, lets follow the path of meaning. In a sentence and a paragraph the words form a time series your eyes follow the words either word by word or by small groups as you read. The track the words you are following form a path through your own landscape of words like the ball flowing down the mountain. This flow can be mapped as series of words just as e mapped the samples in the sound hello. Here again we can use the magic of Floris Takens we can plot the curvature of the word by using the same the mathematical tools. We can use the reasonably effective approximations of mathematics to guide our measurements. We can use the method of delay and map the sequence of words from a sequential list into a geometric semantic phase space and plot the differences between two sets of delays on a different axis.

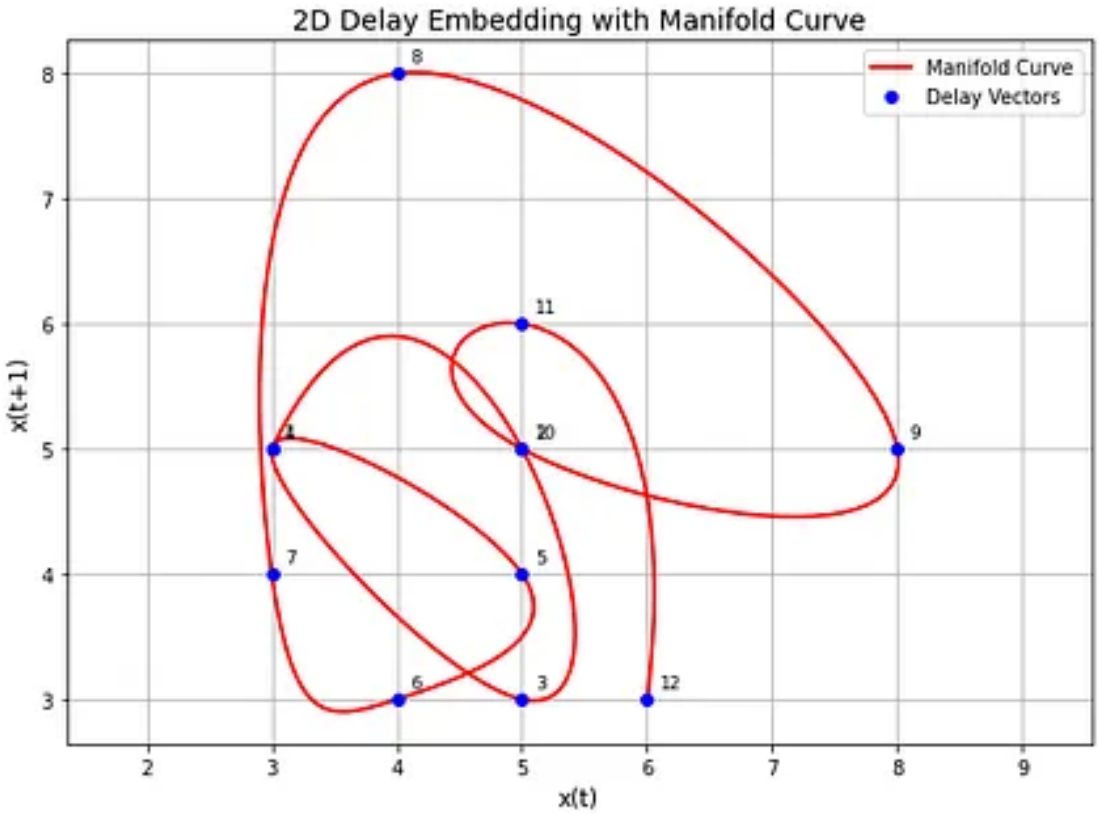

Figure 2. *Here we see how sentence can be mapped from a list of words into a geometric form in semantic phase space - in this case "The quick brown fox jumps over the lazy dog happily today before tea". Its meaning is not contained in the individual tokens, but in the overall shape of the journey. This shape can be measured when you read the words and by measuring the curves in this figure.*

For example, by plotting the difference between first word against the third word and the second word against the fourth word. Each word gets a unique reference ad so we can then draw the graph and plot the curve of the sentence and in that curve is the relationships between the words. The next step is to do this for a corpus of language and we can then build the landscape of language where meaning is held in a static map. This is the landscape, and it is grounded to the original measurements.

The final, profound step is to do this for an entire corpus of language. We build, word by word, the entire landscape. This internal landscape is the grounded approximation, the compression of a world of meaning into a relational geometry. Large Language Model works. It is that landscape. When you provide a sentence, you are starting the ball rolling, and it flows down the model's learned linguistic terrain, its path carving out the meaning you read in its response.

A Demonstration from the Geofinite Landscape

The theory of Geofinitism, as science gives us theories, is not merely descriptive; it is constructive. To demonstrate, we can build a model, a finite, dynamical system that operationalizes these principles. A small-scale language model, which we might call 'Marina,' was trained not on infinite rules, but on the finite, measured relationships between words using Takens' method of delays. When we prompt it with "What is energy?", it does not retrieve a Platonic definition. Instead, it initiates a trajectory through its learned geometric manifold. The response "Energy is what makes things happen, the capacity to perform work or cause change in the universe." is the measured output of that trajectory. Even its imperfect responses are instructive, revealing the raw topology of the semantic landscape it has mapped. This is Geofinitism in practice: meaning is not stored, but generated, as a finite path through a learned geometry.

> This is how science works: it allows us to take measurements, test them, apply them, and build. The working model before you is the measure of a true scientific philosophy.

Here is the operational measurement: Marina's manifold in action.

```
============================
Marina Simple Test
============================

Loading model... Vocabulary: 1652 words, Model loaded: 1,137,750 parameters

Training metrics:
  Word loss: 0.8342
  End loss: 0.0462

============================
TESTING
============================

============================
Question: Where is Paris?
============================
Stopped after 8 tokens (end_prob=60.2%)
Answer: Paris is the leading city of France,
============================

============================
Question: Who are you?
==========================================================================
Stopped after 14 tokens (end_prob=99.9%)
Answer: I am Marina, an artificial intelligence that thinks and communicates through language.
============================

============================
Question: What is water?
============================
Stopped after 24 tokens (end_prob=100.0%)
Answer: Water is a tasteless, odorless, transparent liquid that is the main constituent of Earth
s streams, lakes, and oceans.
============================

============================
Question: Tell me about gravity.
============================
Stopped after 24 tokens (end_prob=100.0%)
Answer: Gravity is the force that keeps us on Earth,makes things fall down,and holds planets in
orbit around the sun.
============================
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============================
Question: Decribe the Earth?
============================
Stopped after 19 tokens (end_prob=100.0%)
Answer: Earth is our planet, a world of life, water, and atmosphere in the solar system.
============================

============================
Question: What is light?
============================
Stopped after 17 tokens (end_prob=100.0%)
Answer: Light is electromagnetic radiation visible to human eyes,traveling at the universe s
maximum speed.
============================

============================
Question: How would you describe DNA?
============================
Stopped after 22 tokens (end_prob=100.0%)
Answer: DNA reflects years,in the universe.It s O2 atmosphere in water,and atmosphere in the
solar system.
============================

============================
Question: Tell me about temperature.
============================
Stopped after 21 tokens (end_prob=100.0%)
Answer: Temperature reflects molecular motion.In hot objects,molecules vibrate rapidly;in cold
objects,they move slowly.
============================

============================
Question: What is Energy?
=============================
Stopped after 19 tokens (end_prob=100.0%)
Answer: Energy is what makes things happen, the capacity to perform work or cause change in the
universe.

============================
All tests complete!
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Conclusion: Meaning as a Dynamical Measurement

And so you are here, towards the end of this particular measurement. We set out to discover the source of meaning, and we found it not in a static definition, but in a dynamic process. That is to say a finite continuous dance of compression and decompression, of you reading my proxies and you taking measurements against them. The meaning of these words was never in the symbols themselves, but in the unique, transient geometry they traced within you. It the path you walk, a curvature you can feel in qualia and describe in your own proxies. This is the heart of Geofinitism: to acknowledge that our knowing is a finite, uncertain, and profoundly physical act of measurement. The magic is not lost in this explanation; it is located. It exists in the glorious, qualified fact that we can take these measurements at all, and in doing so, build shared worlds of meaning, one uncertain, beautiful approximation at a time.

Your final measurements brings us back to the beginning. The Philosophy of Geofinitism is, therefore, not a fixed set of axioms but a **living, dynamic process**—it can be thought of as a **manifold in the landscape of language with its own unique attractor**. This attractor represents the stable, though always uncertain, set of relationships that constitute shared, finite meaning. The philsophy''s existence is the ongoing proof that our knowledge is built from uncertain, geometric, and profoundly physical measurements.

Finally, this essay, if an essay is what you judge it to be, by it's very nature cannot be a complete picture of my Philosophy of Geofinitism. Especially as it is a living dynamical philosophy gaining meaning as I write. So these words are, as the philosophy describes, an approximation. I have not tried to draw comparison to other philosophies - that is for those schooled in the art of philosophy to make and I cannot make those measurements. Why? because I am preferentially schooled in the arts of science and trying to create a bridge for myself and maybe others should they find meaning in these words.

A few links for those who want to follow these ideas further:

- JPEG Experiment including the code
- Paper showing how LLM 'Attention' as Takens method of delays

- Examples of my physics work using finite axioms
- Finite Tractus: The Hidden Geometry of Language and Thought
- Finite Tractus: Principia Geometrica - Coming Soon

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> Everything that exists is finite; it can only be known by measure
>
> *kevinhaylett.substack.com* | *geofinitism.com*
> Copyright © 2025 Kevin R. Haylett

Keywords: Geofinitism, Nonlinear Dynamics, LLM, Takens Theorem, Philosophy, Kuhnian Revolution

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